Despicable Machine



One in a Minion

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FTC Team

#6200

2025-2026

Our Team



Bello!

We are Despicable Machine, a 14th year FTC team from Barrington High School.



Mission Statement:

This season, **we strived to create impact on a large scale by inspiring people all over the world to take on STEM and FIRST**, both at a professional and community level. FIRST has become more than just a robot on a field for our team — it's become a way for us to effect meaningful change into the lives of people and to do our part in **creating the next generation of engineers, coders, leaders, and visionaries**. Implementing the principles of "Gracious Professionalism," we aim to learn, share our knowledge, and inspire others to do the same.

Season Performance Highlights

- **3rd place overall** @Meet 3
- **2nd highest score** @Meet 3
- **7 individual mentors**
- **83 FLL teams mentored**
- **160 robotics kits donated**
- **$8,000 raised**

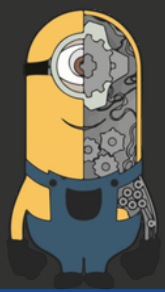
Specfic Season Goals	How we accomplished them
Raise enough money to fund our outreach and design aspirations	We cold emailed 126 companies, presented to 25, and **gained sponsorships from 11**
Improve the performance of our robot by increasing localization accuracy during autonomous	We implemented unique code such as **4 wheel odometry** and the use of **Limelight vision**
Use the design process and custom made parts to enhance the build	We invested in **3D printing materials** and contacted mentors to learn more about design principles
Spread STEM access to groups with limited or no access	We started most of our outreach over the summer, adhering to strict deadlines and focusing on goals

Gracious Professionalism

Our team strongly emphasized embodying this core value of FIRST this season. In G.E.A.R.S., **we educated all recruits on the importance of gracious professionalism** and how to implement it. In addition, we **collaborated with other FIRST teams** this season many times, both as mentors and mentees. By centering our team around gracious professionalism, we represent FIRST at its core.

Meet the Team!





Together, we make Team #6200, **Despicable Machine!** You could say a team like this is ***One in a Minion!***

6200's plan for sustainability

Managing Our Risks

- Ensure financial stability for **future seasons**
- Establish a team structure that **outlasts current membership**
- Strategically plan and train for **long-term success**

Season Plan



- Train new members through **GEARS**
- **Delegate members** to different team roles
- Recruit **future members** from outreach
- Fundraise **$5k**
- Increase number of **returning sponsors**
- Prepare novices to **replace** senior leaders
- Develop each members skillset with a **personal mentor**

Tracking Progress

- Subteam meetings **every week**
- **Daily** engineering notebook entries
- New members progress evaluated through **GEARS**
- Team finances logged and checked **each week**
- Follow-up **newsletter with sponsors** monthly
- Track outreach engagement and participation statistics

Financial Plan

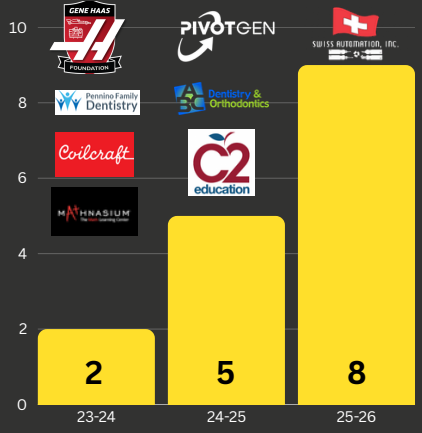
$8,000 Raised from 11 sponsors!

How we budgeted:

- Established a **projected budget at the start of season** for anticipated expenses
- Informed based off **past seasons**
- Tracking expenditures **across seasons** helped us evaluate the effectiveness of our fundraising

Money	Projected Expense	Money	Notes
-	Based on average of previous 2 years		Returning Sponsor
$2,000	Robot parts	$1,000	PivotGen
$600	Presentation-related materials	$1,000	Swiss Automation
$236	Website costs	$500	Coilcraft
$2,000	Outreach materials (e.g. robotics supplies)	$400	**BHS Concessions**
$174	Tshirts	$100	Pennino Dentistry
$1,000	Team expenses and dues	$250	ABC Denistry
$500	Game field	-$1,000	Team Expenses
~$1500	Rollover funds for next year!	-$500	Game Pieces & Field
		-$174	Team Tshirts
		$1,000	**Main Street Smiles**
		$250	C2 Education
		-$32	Wix Website Costs
		-$204	Wix Website Costs
		$2,000	Gene Hass Foundation
		-$208	Presentation Trifold
		$250	**Barrington Education Foundation**
		$250	Barrington Mathnasium
		$1,000	**Oamic Ingredients**

Number of returning sponsors



How we fundraised:

Retaining sponsors

1. Progress updates
2. Invitations to outreach and competitions
3. Robot demonstrations

Recruiting sponsors

1. Cold email/call
2. Connections from outreach events
3. LinkedIn or personal connections

6200 Sustains: Future teams

Recruiting new members

Middle School Outreach

- Introduced local middle schoolers to FTC Robotics
- Interactive, hands-on robot demonstration

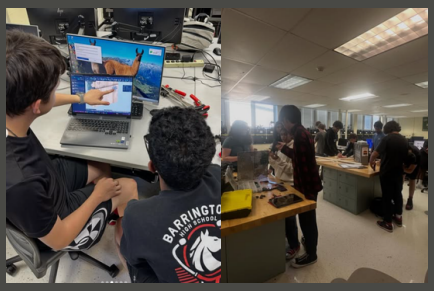
20+ Students joining us next season for club and/or competition!



A recruitment pipeline to introduce students **without prior robotics experience** to FIRST programs and **generate robotics interest.**

High School Clubs

Engineering Club



Robotics Club



Gir1C0de Club



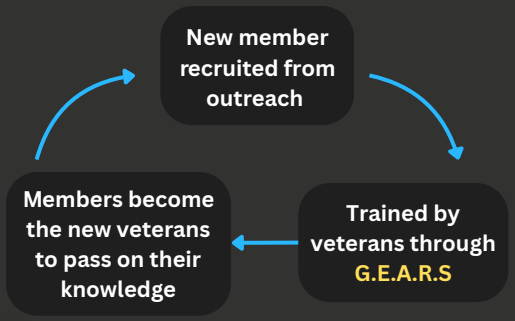
Two of our team members were recruited from robotics club!

Training new members

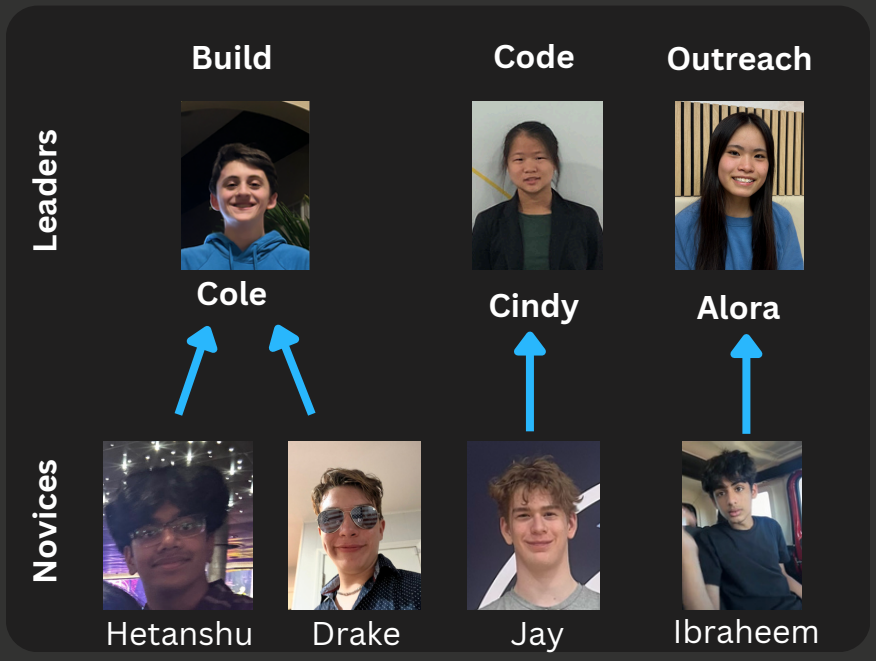
G.E.A.R.S. Program

- A **one-on-one mentorship program** designed by our team to teach **essential FTC skills** to recruits (building, coding, CAD, and portfolio)
- Partners **new and veteran members**





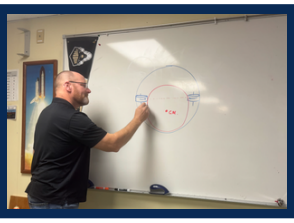
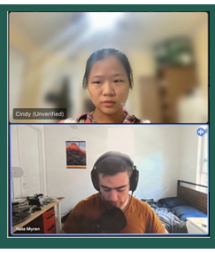
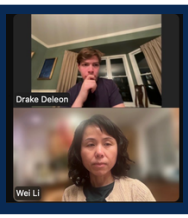
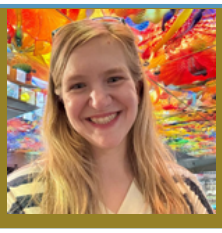
Our Team Structure



6200 Connects: Professional Outreach

Mentoring new skills

Biweekly meetings with mentor

Team Member and Target Skill	Mentor	The Result
Cole - Lead Builder Applying physics principles to robot desgin	 **Matthew Gelon** Civil Engineer Physics Educator	• Improved understanding of the physics of launching systems • Modified launcher for more efficient energy transfer and increased scoring consistency (see page 9)
Cindy - Lead Coder Improving autonomous with computer vision	 **Nate Myren** Software Engineer at Google	• Learned more about vision pipelines and localization with AprilTags • Integrated a Limelight vision-based position adjustment into our autonomous, increased shooting accuracy in auto & TeleOp
Alora - Lead Outreach Developing a robotics curriculum for K-8 audiences	 **Joseph Kabbes** Director of STEM outreach at FSU Civil Engineer	• Learned how to transfer complicated, intimidating science topics into simple terms for K-8 audiences • Improved robotics workshops with more engaging and instructive activities like competitions!
Hetanshu - Build Use testing to improve robot functionality	 **Karen Kabbes** Civil and Environmental Engineer	• Determined points of failure in our initial intake system • Learned to use the design process to iterate and fix critical issues in the intake (see page 8)
Drake - Build Improving robot design & iteration with CAD and math	 **Wei Li** Senior Systems Engineer at Honeywell	• Learned how to translate desired robot goals into CAD models • Led to more precise robot design and accuracy in 3D modeled parts
Jay - Coder Using encoder feedback for more auto accuracy	 **Lisa Han** Project Manager at Microsoft	• Learned how to use the engineering design process in software • Studied PID control loops • Improved autonomous accuracy with a 4-odometry wheel system
Ibraheem - Outreach Planning engaging community outreach events	 **Kim Salada** BCGT Director Former senior program manager at Microsoft	• Learned how to develop engaging and interactive outreach programs • Created outreach events with more community involvement ◦ Toy drive, mentorship programs

6200 Connects: Professional Outreach

Reaching Out to the Engineering Community

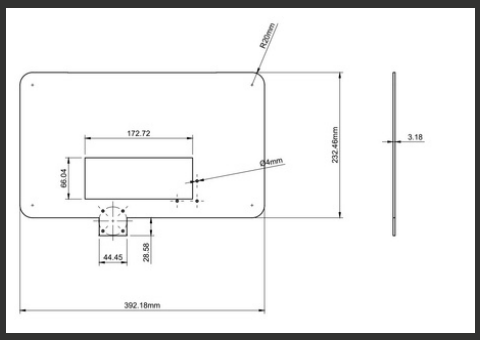
Learning about engineering in the workforce

- Toured facilities of professional STEM companies to experience how robotics and engineering principles can be **applied to real world problems**
  - **Swiss Automation**, a local parts engineering company
  - **Kabbes Engineering**, a civil and environmental engineering company.





Our CAD file



Swiss Automation made side plates

- Swiss Automation CEO **suggested side plates to improve robot sturdiness and defense** during our visit
- **Taught us** about machining and CAD
- Worked directly with **Swiss Automation** to manufacture **custom side plates**
- Added space for a **magnetic team number holder** for aesthetic design
- Helps **contain wires** and critical **internal components**

6200 Reaches: Involving Community

ISBE CTE Conference

IMPACT: 350 educators

- Spoke about the **importance of robotics and STEM education** to 350+ educators
- Discussed our FIRST journey and ways of increaisng robotics participation in schools



STEM Toy Drive (400+ toys)

IMPACT: 400 students





- Collected and donated **400+ toys** (estimated value *$6k+*) to **children at Alexian Brothers Hospital and ICNA Relief Refugee Center**
- For children hospitalized over the holidays and refugees in underfunded schools

6200 Reaches: Embracing FIRST

Sent 160 Robotics Kits to Underserved Schools

IMPACT: 2,000 students

- Targeted schools **in Zimbabwe** without resources to establish robotics programs
- Used by **thousands of middle schoolers** as their **first exposure to robotics**





Created and Distributed 8 Lesson Curriculum

- Beyond providing the resources we **provided instruction**
- Students had **very little to no background** in STEM
- Provided lessons covering robotics basics and further STEM involvement

Partner Schools:

- **USAP Community School**
  - Serves high-achieving low-income students and currently has 16 students
- **Uncommon**
  - Provide STEM education to students (mostly first time coders) with 3000+ students

BCGT Robotics Workshops

IMPACT: 70 students





- Hosted **hands-on workshops** introducing students from K-8th grade to FIRST robotics
- Participants from over **20 different schools**
- Taught programming, building, engineering design, and gracious professionalism

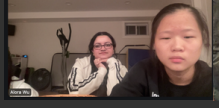
FLL Mentorship (83 teams)

IMPACT: 750 students

- Hosted **5-week workshop** for **83 FLL teams** from 18 states/regions, 31 being first-time coaches
- Individually **mentored 8 FLL teams in person and online** across three months, with **5 winning awards**





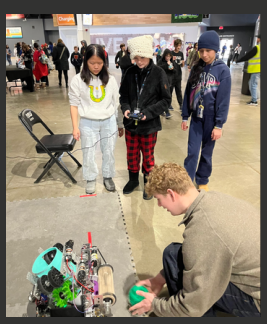




Windy City Bulls

IMPACT: 5,950 people

- Demonstrated our robot interactively to thousands of elementary and middle schoolers
- Spread the values of FIRST & hands-on STEM





9520+ students interested in FIRST

"The passion these students have for spreading STEM is beyond inspiring and has really shined through this season." *- Mrs. Dowdy, Elem. Teacher*

6200 Thinks: Design Process

Throughout the season, we used **the design process** to build/improve our robot.

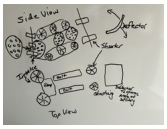
1.Define the Problem

Set specific goals we wanted our robot to achieve:



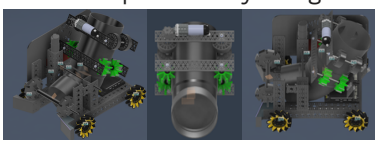
2.Generate Concepts

Worked together as a team to brainstorm solutions



3.Develop Solution

Used Fusion 360 and OnShape to CAD preliminary designs



4.Test Prototype

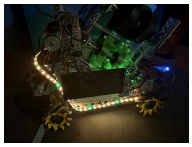
Created prototypes of best solutions and test on the field



5.Present Final Solution

Build robot using off-the-shelf and custom machined parts





6.Evaluate

Test robot and locate areas of improvement after each meet

- Whole team meetings after each meet to reflect

Dates: 11/14/2024
Attendees: Justin, Cindy, Aidan, Alora, Virginia, Zarah
Goals for Meetings:
- ☐ Reflect on Meet 1
- ☐ Find areas of improvement/plan

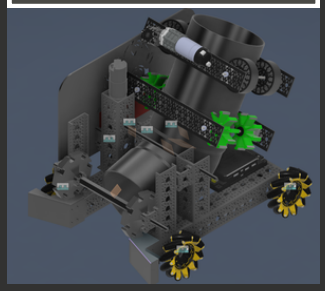
At the beginning of each season and after each meet, we **met as a team to reflect** on our **strengths, weaknesses, and lessons learned** during matches, **set goals**, and create a timeline for implementation to improve our robot design moving forward. **Math and physics and extensive testing** heavily guided our design and iterations this season.

An Overview of Our Robot:

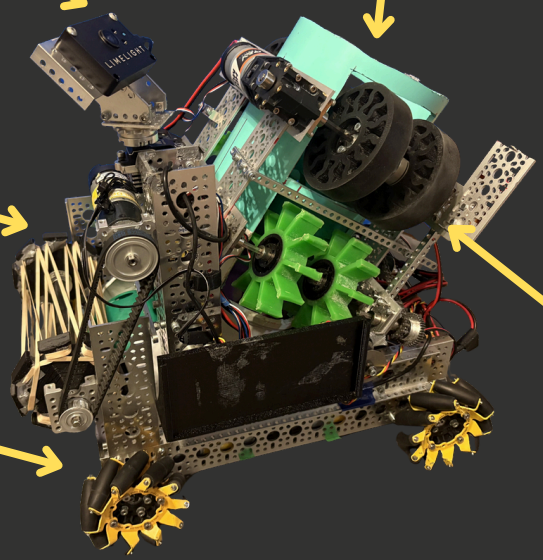
Limelight on Servo
Bridging software and hardware control solutions

PVC Tube
To ensure consistency in delivery

CAD
Was always the first step in our process to model our design



Protruding rubber band intake
Quickly captures wide area of artifacts ("touch it, own it", page 8)



Delivery system
Extensively tested to find optimal gecko wheel/motor torque combination (see page 10)

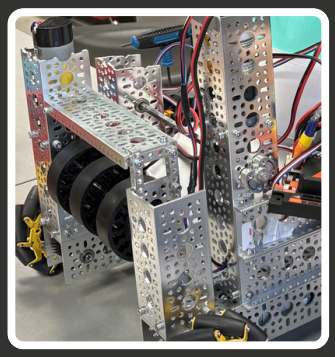
Mecanum drivetrain
To support defensive movement

6200 Designs/Innovates: Our Robot

A Deeper Dive Into Our Robot: Intake System

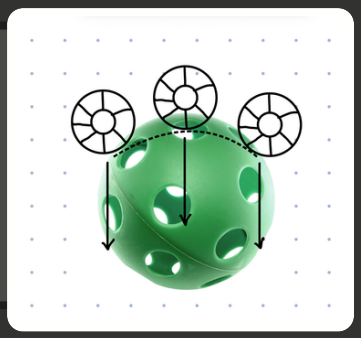
GOAL: Have rapid and easy intake

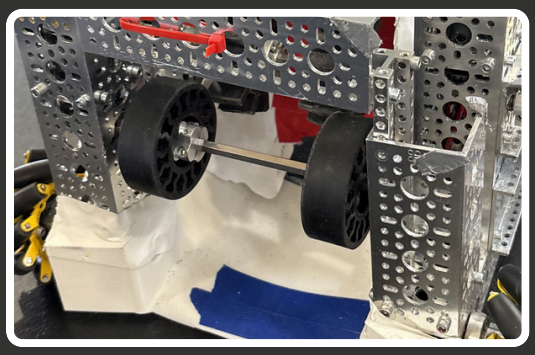
"Touch it, Own it."



Iteration 1

Our first iteration used three 48mm gecko wheels to apply compression to the artifact and intake it. However, we noticed that the applied **compression was too high**, especially in the middle wheel, meaning artifacts had trouble moving into the robot.



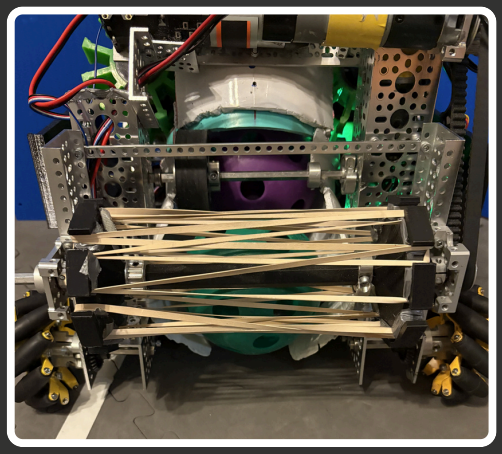


Iteration 2

- Removed one of the gecko wheels to **decrease the compression** on the artifacts.
- Experimented with the spacing of the two wheels to find the optimal width for artifact intake, applying just enough compression to bring the ball in.

Result

- **Very small margin of error** for intake. Artifacts must be perfectly aligned, significantly increasing the amount of time to intake.
- Artifacts close to the robot **get stuck** in the plastic edge before the start of the tube and get pushed away before contacting the intake.



Final Iteration

Protruding Rubber Band Intake

- Extended the intake outside of the robot frame
- **Easier and quicker** to intake artifacts
- Artifacts no longer get stuck at the tube entrance
- Wider intake increases capture area, allowing artifacts to enter from **more positions and angles**
- **Margin of error increases tremendously**

Risk Management

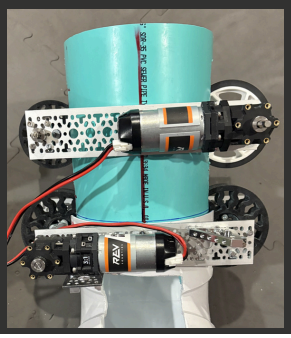
One risk we noticed throughout practice was that the rubber bands would **occasionally snap**. To mitigate this possibility, we **double-laced the rubber bands** so that if one broke, a backup would allow the intake to remain functional.

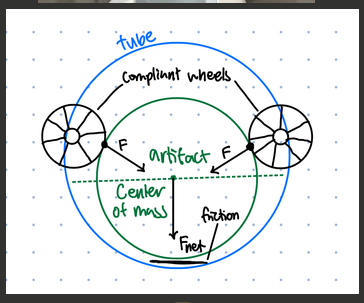
A Deeper Dive Into Our Robot: Delivery System

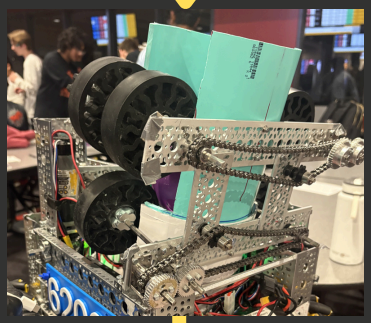
GOAL: Achieve consistent, accurate shooting from both launch zones

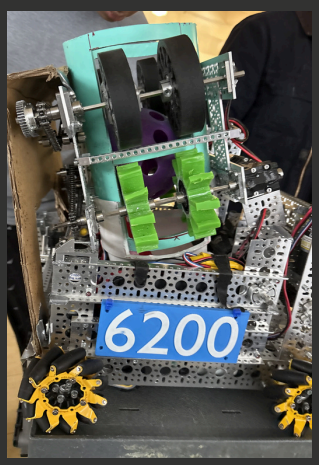
Dual-Horizontal Compliant Wheel Flywheel Shooter

Maximizing horizontal shooting consistency and control of vertical distance









Iteration 1

- **Single compliant wheel** on either side of the artifact.
- We experimented with multiple different configurations, but found that **no design provided the sufficient power needed** to consistently cleared the goal.

Advice from Mr. Gelon

To tackle this issue, we talked with our build mentor Mr. Gelon, an **engineer and physics teacher**. He suggested:

- Wheels might not be contacting the artifact at the center of mass, **driving the artifact further into the PVC tube** and **increasing friction**, rather than propelling it into the air.



Iteration 2

- **2 compliant wheels** on either side in the final shooting stage
- The second wheel:
  - **Centers the artifact** in the shooting tube, moving it away from the sides and **reducing friction**
  - **Increases contact surface area** with the artifact, allowing **more energy** to be transferred.

This new design allowed us to **consistently clear the goal.** However, we noticed that we still **did not have the sufficient power** to shoot from the far launch zone.

Final Iteration

- **Added 2 wheels** on either side of the 3rd shooting stage
- **Adjusted angle of tube** to achieve greater horizontal distance
- Switched to **cut-up compliant wheels** compared to smooth ones
  - **Increases grip and the contact surface area** with the artifact
  - Opted against switching the fourth stage of wheels to cut-up gecko wheels as testing showed that this increased inconsistency in shooting distances
  - Both these adjustments **increased energy transfer** to the artifact and helped us with shooting from the far launch zone.

A Deeper Dive Into Our Robot: Delivery System

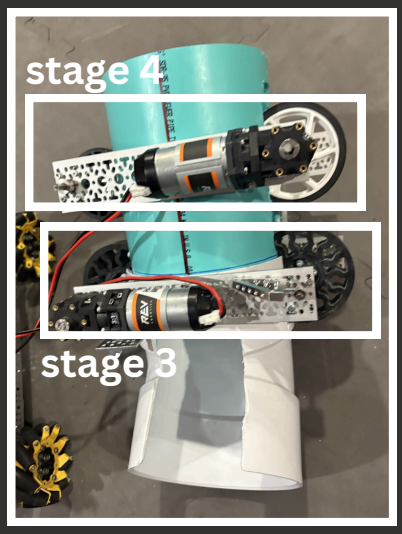
PVC Pipe Shooting Delivery Mechanism

Rigid Cannon

- Consistently ensures that the artifacts are launching **in the same direction**
- Increasing **replicability** and **accuracy** of our shooting

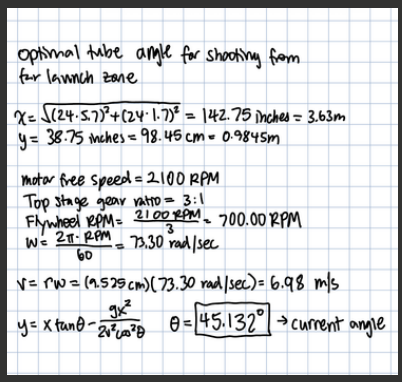
Human Player Intake

- **Doubles as a method** for **human player intake** when the top three stages of wheels are run in reverse, allowing us to **feed in artifacts**.
- Helps us to **easily build patterns** in endgame



Easily Adjustable with Modular Design

- PVC pipe can be **easily removed** from the rest of our robot
  - Allowed for easy testing during early development
  - In the later stages, helped us **easily adjust the angle of** the artifact **shooting** without needing to change the rest of the robot



***Above**: Physics computations to determine the optimal angle of the tube to shoot to the goals from the far launch zone based on our motor speeds.*

4 of 24 Total Motor Torque Testing Trials & Results

4th stage motor τ	3rd stage motor τ	Distance shot	Cost/Benefit analysis
1:1	15:1	Very short	High τ in 3rd stage reduces slipping, low τ in 4th stage → ball almost never clears goal
3:1	**9:1**	**Far**	**Medium τ in 3rd and 4th provides enough grip and speed**
4:1	12:1	Medium	Too high τ in 3rd stage doesn't give enough speed to get artifact to 4th stage
5:1	9:1	Medium	Lower maximum speed in 4th stage, better energy transfer in 3rd stage

While testing our delivery design, we learned that the **balance between torque and speed** was very important to achieve **ideal shooting.** Higher torque helped launch the artifact out, while higher speed did so to a greater distance. We **tested multiple combinations** (24 in total!) of 3rd and 4th stage motor torques (τ) to determine the optimal setup. We used the REV UltraPlanetary HD Hex Motors to **easily switch the gear ratio**. We also tested compliant wheel combinations for our shooter with varying hardnesses (i.e. 30A).

Compliant Wheel Testing Results (3 of 16 tests)

Wheel 1	Wheel 2	Analysis
3.54", 65A	3.78", 30A	Lopsided, curved shot
3.78", 30A	**3.78", 30A**	**Straight, accurate shots**
2.25", 35A	2.25", 35A	Too low velocity

Lessons Learned

We embraced the importance of **extensive testing** to back our ideas and find new solutions, **rapid prototyping and continued iteration**, and supporting design choices with **math and physics.**

6200 Controls: Our Software

Our Robot Can...

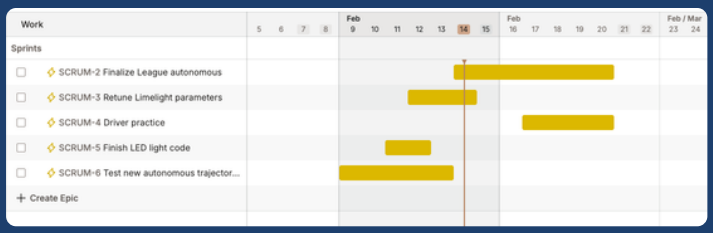
1. Shoot three preloaded artifacts into the goal **(97% accuracy)**
2. Intake & shoot 6 to 9 additional artifacts (varying based on starting location) with Limelight-aiming, with multiple Opmodes to assist partners **(95% accuracy)**
4. Use **preprogrammed buttons and sensors** to **assist during driver control**

Preliminary Goals/Implementation

GOAL: Give everyone a chance to code

With 2 new members interested in coding, we focused on increasing collaboration in our workflow.

1. Created an **FTC Code Guide** to teach new members through the GEARS program.
2. Used **a Github repository with multiple branches** for version control and to facilitate **collaborative coding**.
3. Utilized **Jira to improve agile workflow**



GOAL: Enhance code workflow

Improving our programming tools helped us be more organized and efficient.

We used git for effective version control throughout the year.

We switched to Android's Wireless ADB for code upload, saving time.

We used a USB Wi-Fi adapter to connect to the Control Hub and use the internet simultaneously.

Throughout the season, we used the **engineering design process** to make improvements upon our code. At the beginning of the year and after each meet, we met as a team to reflect upon and plan **software goals** around the robot functionality we wanted to achieve. We made **iterations based on extensive testing** to improve our designs.

Defining the problem/goal	Rapid prototype of control/software solutions	Testing of solution
What do we want to accomplish? Increased accuracy, speed, etc.	What approaches do we take? What control elements & software do we use?	How effective is our solution? Does it work reliably and accurately in matches?

improve solution

A Deeper Dive Into Our Code: Autonomous

GOAL: Accurate drive motor control & localization to support consistent shooting and pathing

1 Control Loops for Drive Motors

To **minimize error** during autonomous and with automated portions of driver control, we used control loops with proportional, integral, and derivative **(PID)** terms. For our motors, we used the following feedforward control formula to create **custom PID control algorithms.**

$$f(t) = K_v \times velocity + K_a \times acceleration$$



Trapezoidal motion profile for increased accuracy over longer distances

However, we noticed that control loops alone **didn't provide the accuracy needed** to make reliable shots into the goal, as using the drive motor's internal counts for the position input yielded inconsistent results that didn't account for error due to friction.

2 Custom 4-Wheel Odometry Design

To achieve **higher localization accuracy**, we chose to use odometry pods to track our location along the field. While most teams only use 3 odometry pods, we added a **4th dead wheel to gather additional data.** This improved accuracy allowed us to navigate complex splines during auto, saving time during intake & shooting and increasing scoring.

Iteration #1

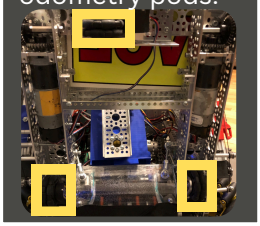
Custom-designed odometry pods with 3 REV Through-Bore Encoders.



Inconsistent ground contact & revolution counting

Iteration #2

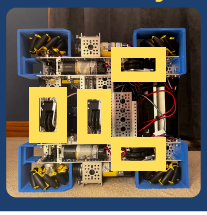
3 goBILDA odometry pods.



*High-resolution encoder gave **precise position information,** Spring loading **improved ground contact***

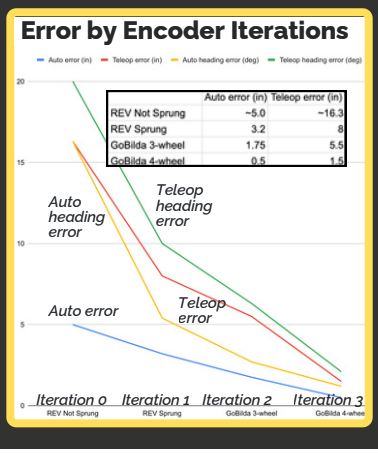
Iteration #3

Custom 4-wheel odometry setup.



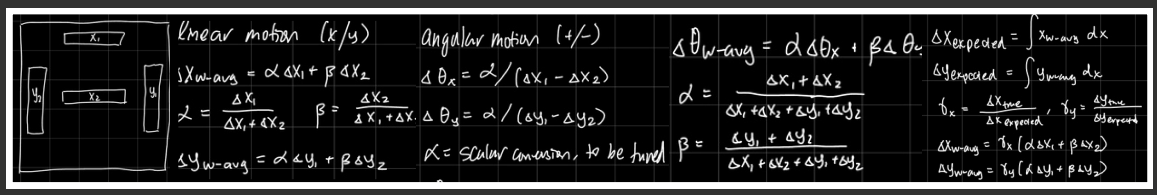
Most teams use only 3 dead wheels.

*Redundant data from the extra wheel **improved localization 3x***



	Auto error (in)	Teleop error (in)
REV Not Sprung	~5.0	~16.3
REV Sprung	3.2	8
GoBilda 3-wheel	1.75	5.5
GoBilda 4-wheel	0.5	1.5

Adaptive Weighted Average Algorithm for 4 encoders



***Takes into account error from encoders**, compares with other localization methods (i.e. Limelight).*

Risk Management

Encoders in odometry pods can sometimes break, greatly skewing localization data. We account for the risk of inaccurate encoders within the parallel wheels by checking the difference in their readings, and using just one of them if the difference is too large. Taking a weighted average also allows us to **minimize individual encoder errors.**

$$x(t) = x_0 + v_x t, y(t) = y_0 + v_y t$$

Above: spline formula. We also used **temporal markers** and **async functions** to save time thorugh autonomous.

A Deeper Dive Into Our Code: Computer Vision - Limelight

GOAL: Achieve consistent and rapid shooting by further increasing localization accuracy

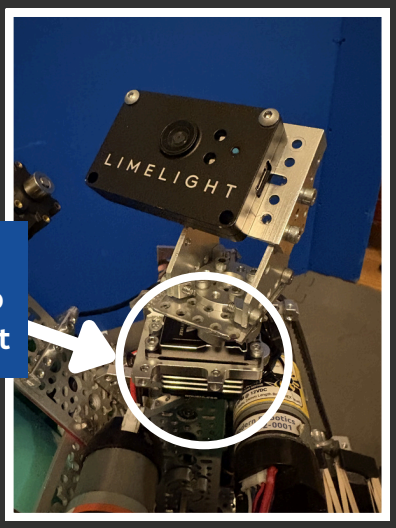
AUTO PROBLEM!

Inconsistencies with initial robot setup → Throws off rest of auto trajectories

SOLUTION

Limelight recalibrates position throughout auto based on external feedback with AprilTag

We used the **Limelight Smart Camera** to perform **vision-based relocalization** each time we shoot during autonomous. We obtain the robot's pose relative to the AprilTag on the goal and adjust the robot's positioning accordingly to make an **accurate and consistent shot.**



TELEOP PROBLEM!

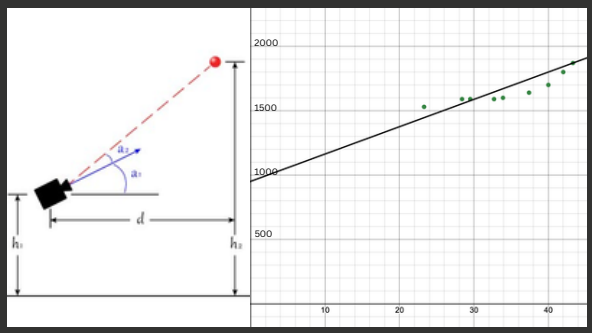
Static shooting velocity only works for one position → Slow cycle times in TeleOp

During our first 2 competitions, using a singular static shooting speed meant our driver needed to line up to a **very specific location** on the field to score an artifact, resulting in **inconsistent shooting** and **slow cycle times**. To solve this, we used a Limelight camera to track AprilTags and determine the robot's relative position to the goals.

SOLUTION

Limelight automatically changes shooting velocity based on distance to AprilTag

Using the **distance and angle data from the Limelight**, we tuned a regression function to **dynamically vary the shooting speed** of the shooting motors based on how far from the goal we are. We also created a drive control button to **automatically adjust** the robot to **the optimal position** and angle for shooting, **removing driver error** during delivery.



$$distance = (h_2 - h_1)/\tan(a_1 + a_2)$$
$$velocity = 21.2 \cdot (distance - 40) + 1800$$

TELEOP PROBLEM!

Limited Limelight FOV slows down driver during shooting

However, the Limelight's narrow field of view meant the robot often couldn't see the AprilTag. Thus, we combined hardware and software design to **mount the Limelight on a servo**, which rotates to ensure a constant line of sight to the AprilTag, **increasing the effective FOV**. This allows us to score artifacts from **anywhere on the field** with the click of just one button, removing any fine-tuning adjustments needed and increasing scoring.

SOLUTION

Limelight on servo increases FOV Decreased cycle times!

A Deeper Dive Into Our Code: Automation in TeleOp

GOAL: Use control solutions to make TeleOp scoring efficient and consistent

Throughout the season, we iterated upon and altered ideas to improve TeleOp performance based on feedback from our driver and strategy teams.

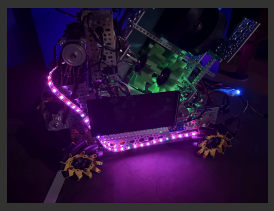
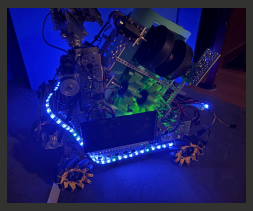
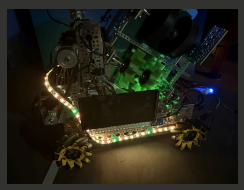
1 Relaying Info Via LED Lights

- Limelight Computer Vision
- Servo for LimeLight
- REV Distance sensor
- Java System Time

→ Determine when the robot is within shooting distance

→ Automatically adjust shooting speed based on distance to AprilTag

→ *Relays that info to the driver via LED lights attached to the robot*

LED Color Key

- Ready to shoot
- Ready to shoot, unsafe distance (too close or too far)
- Out of shooting range
- 3 samples intaked
- Endgame warnings

We wanted to **communicate important information to the driver to streamline scoring.**

2 Holonomic Drivetrain

Based on this year's defense-heavy game, we elected to use a holonomic drivetrain with mecanum wheels, which allows us to move in any direction, **improving defense and scoring.**

Vector equations for our drivetrain

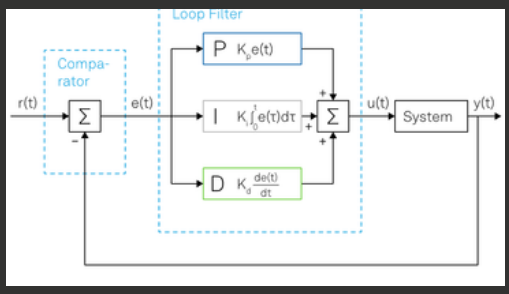
$$x_2 = \cos\beta x_1 - \sin\beta y_1$$
$$y_2 = \sin\beta x_1 + \cos\beta y_1$$

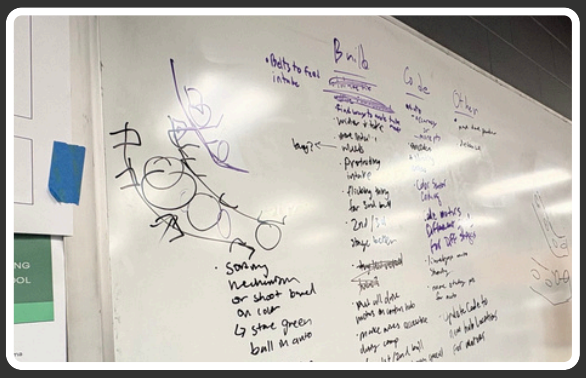
PID equation for motor control

$$f(t) = K_p e(t) + K_i \int_0^t e(t)dt + K_d \frac{de(t)}{dt}$$

3 Custom PID Loops for Motor Control

Automatically adjusting shooting speed based on the distance from the goal meant that we needed precise control of the speed of our 4th stage shooting motors. To achieve this, we created **custom PID algorithms** that use the motor encoders as feedback.





Lessons Learned

We learned the importance of **constant iteration,** the importance of **clean and well-documented code,** the functionality of **sensors** to better performance. The principles helped us come up with **creative solutions** to increase our **accuracy, consistency, and speed.**

6200 Season Highlights

Reach/Sustain

- Sent robotics kits to **2000 kids**
- Hosted **5-week** FLL workshop
- Hosted BCTG intro **robotics workshop**
- Fundraised **400 STEM toys**
- Implemented **GEARS**
- **Recruited future members** from clubs/middle school
- Financially stable season

Connect

- Each member developed **target skills** with their **personal mentor**
- Learned from local engineering professionals and **toured several companies**
- Worked directly with **Swiss Automations** to manufacture custom side plates

FIRST TECH CHALLENGE

Design/Innovate

- Protruding rubber band **intake**
- Dual-Horizontal Compliant Wheel Flywheel **Shooter**
- PVC Pipe **delivery** mechanism
- Extensive testing and multiple iterations made in the **design process**

Control

- Utilized **engineering design process** to improve software
- Control loops for drive motors
- Custom **4-wheel odometry system**
- **Limelight aided computer vision** with auto adjustments
- **LED** light TeleOp signals
- **Custom** PID loops

Lessons Learned

- Increased **build efficiency with CAD** and math/physics calculations
- More accurate robot mechanics when physics principles are applied, taught by Mr. Gelon
- **Secured more returning sponsorships** with a strong pitch and involvement with companies
- Greater team productivity with implementation of GEARS
- Achieved greater autonomous accuracy with more control software

Future goals

- Spread FIRST to more communities by **expanding robotics curriculum**
- **Further develop GEARS**, the team training program
- Enhance robot performance with new technologies
- **Build strong relationships** with local engineering professionals
- Increase number of returning sponsors

Plans to Achieve Them

- **Engage local community members** and new schools to spread FIRST
- Implement a transition program for middle school robotics
- **Research machine learning and machining techniques**
- Reach out to nearby engineers to gain more technical insight and sponsorships

We are so grateful...

to be able to share our season with you...



to inspire others to take on challenges....



and to collaborate with people around the world!



And if there's one thing we learned this season,

in order to make an impact, someone has to do it...

FIRST®